

Module 5

Applied Mathematics for Machine Learning

Technical Workouts

1. Understand Linear Algebra Fundamentals
 - a. Scalars
 - b. Vectors
 - c. Matrices
 - d. Matrix operations
 - e. Matrix multiplication
2. Understand Vector Mathematics
 - a. Vector representation
 - b. Magnitude
 - c. Direction
 - d. Dot product
 - e. Similarity concepts
3. Understand Matrix Applications in ML
 - a. Data representation
 - b. Feature matrices
 - c. Transformations
 - d. Matrix-based computations
 - e. Real-world ML applications
4. Understand Calculus Intuition
 - a. Functions and rates of change
 - b. Slopes
 - c. Derivatives (conceptual understanding)
 - d. Optimization intuition
 - e. Why calculus matters in ML
5. Understand Optimization Fundamentals
 - a. Cost functions
 - b. Error minimization
 - c. Gradient intuition
 - d. Optimization workflows
 - e. Learning through iteration

6. Understand ML Mathematics in Practice

- a. Linear Regression intuition
- b. Similarity calculations
- c. Distance measurements
- d. Feature transformations
- e. Mathematical reasoning in ML

7. Thinking with AI

- a. Use AI for:
 - i. Formula explanations
 - ii. Matrix operation understanding
 - iii. Optimization visualization
 - iv. Validate mathematical reasoning
 - v. Verify generated solutions

8. Apply Through Hands-on Tasks

- a. Perform vector and matrix operations
- b. Analyze feature relationships
- c. Visualize optimization concepts
- d. Apply mathematical reasoning to ML problems
- e. Explain how mathematics powers ML models
- f. Interpret matrix-based datasets